AI Teaching Assistants in Asynchronous Discussion Forums: A Mixed-Methods Study of Students' Community of Inquiry Presence

Yi Tao[a], Heting Li[a], Yuyin Lin[a], Jieru Lu[a], and Yingbin Zhang[b*]

[a] School of Information Technology in Education, South China Normal University, No.55 West of Zhongshan Avenue, Guangzhou, Guangdong 510631, China.

[b] Institute of Artificial Intelligence in Education, South China Normal University, No.55 West of Zhongshan Avenue, Guangzhou, Guangdong 510631, China.

Abstract

Grounded in the Community of Inquiry (CoI) framework, this study examined how an artificial intelligence teaching assistant (AI TA) shaped students' perceived and behaviorally enacted CoI presence in asynchronous discussion forums. TA support was examined in a 10-week blended course across the no-TA stage, human-TA stage, and AI-TA stage, with a historical cohort receiving human-TA support included for behavioral comparison. An explanatory mixed-methods design was adopted, combining CoI questionnaires, student-authored forum posts, forum-based reflections, and semi-structured interviews. The results revealed a perception–behavior gap: students reported higher cognitive, social, and teaching presence during the AI-TA stage, but these perceived improvements were not mirrored in forum behaviors. Behavioral cognitive presence increased significantly from the no-TA stage to the human-TA stage but showed no further increase in the AI-TA stage, while behavioral social and teaching presence remained sparse across stages. Qualitative findings suggested that, compared with human TA, students perceived AI TA as advantageous in structured cognitive

support, a low-pressure interaction atmosphere, and timely and accessible guidance, but limited in inquiry flexibility, emotional authenticity, and social authority. These findings highlight the limitations of AI TAs in ADFs: they can support basic cognitive activities, but their capacity to foster deeper inquiry and peer interaction depends on human orchestration and complementary facilitation. Future research should explore human–AI collaborative facilitation models that leverage the strengths of AI TAs while preserving the pedagogical and social roles of human TAs and instructors in asynchronous discussion forums.

Keywords: artificial intelligence teaching assistant; asynchronous discussion forums; Community of Inquiry; pedagogical agents; cognitive presence

Introduction

Asynchronous discussion forums (ADFs) support reflection, peer interaction, collaborative knowledge construction, and critical thinking in online and blended learning (Galikyan & Admiraal, 2019; Hull & Saxon, 2009; Miyashita & Wark, 2024; Robbins & Fairbank, 2023). However, instructors have limited capacity to provide timely, meaningful responses to large volumes of posts, and delays in peer interaction can further weaken feedback. Existing remedies—including teaching assistants, optimized instructor–student ratios, and reply-priority systems—remain dependent on scarce human resources (Irish et al., 2023; Parks-Stamm et al., 2017; Yang, 2008). A scalable approach is therefore needed to sustain facilitation in ADFs.

Pedagogical agents (PAs) offer one such approach by combining course-specific knowledge with theory-informed interaction logic. AI-powered agents such as Khanmigo and Jill Watson provide Socratic guidance, answer course-related questions, and support engagement with course

materials (Design Intelligence Lab, n.d.; Khan Academy, n.d.; Taneja et al., 2024). However, their use has centered largely on tutoring and course support rather than sustained peer-discussion facilitation.

The use of PAs as embedded facilitators in ADFs remains underexplored. Existing PAs have primarily supported individual tutoring and platform-based learning (Apoki et al., 2022; Martha & Santoso, 2019), with limited attention to ongoing peer discussion. This study therefore designed a PA that functions as a teaching assistant within ADFs. Because cognitive presence (CP), social presence (SP), and teaching presence (TP) are central to meaningful online learning (Martin et al., 2022), the CoI framework guided the AI TA's design and evaluation. An explanatory mixed-methods design examined students' self-reported and behaviorally enacted CoI presence and their perceptions of the AI TA across TA-support conditions.

Literature Review

CoI Presence in Online Discussion Forums

The CoI framework argues that cognitive, social, and teaching presences interactively support meaningful learning processes (Garrison et al., 1999; see Fig. 1). In online education, it provides theoretical guidance for the design of learning environments and the evaluation of learning experiences (Akyol & Garrison, 2008). Higher perceived levels of the three presences have been associated with better course grades, student experience, perceived learning, and satisfaction (Rockinson-Szapkiw et al., 2016; Stenbom, 2018).

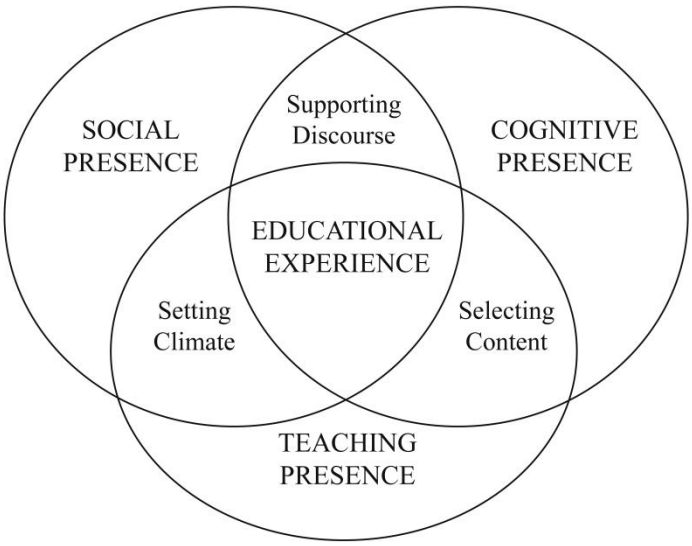


Fig. 1 Community of Inquiry framework. Adapted from Garrison et al. (1999).

.CP refers to the degree to which learners "collaboratively construct meaning through sustained discourse within a CoI" (Garrison et al., 2001). It is commonly analyzed through the Practical Inquiry Model, including triggering events, exploration, integration, and resolution (Garrison et al., 1999). A persistent challenge in asynchronous discussions is that cognitive activity often stalls at the exploration phase, rarely reaching higher-order integration or resolution (Garrison & Arbaugh, 2007; Naeem & Khan, 2019). This suggests that merely providing an online space for discussion is insufficient; students need strategic prompts, questioning, synthesis, and guidance to move from information exchange toward deeper inquiry (Darabi et al., 2011; Koh et al., 2010).

SP is defined as the "capacity of participants in a CoI to project themselves socially and emotionally as 'real' people" (Garrison et al., 2010). It serves as the foundation for a collaborative learning community by supporting affective expression, open communication, and group cohesion. In ADFs, SP is especially important because students interact without immediate

face-to-face cues and must rely on textual communication to establish interpersonal connection, trust, and group belonging. However, sustaining SP remains challenging when peer responses are delayed, participation is uneven, or students perceive the forum as a task-completion space rather than a learning community.

TP involves the "design, facilitation, and direction of cognitive and social processes" to achieve meaningful learning outcomes (Anderson et al., 2001). It is enacted through instructional design, discourse facilitation, direct instruction, and activities such as clarifying expectations, encouraging participation, maintaining discourse, asking probing questions, and providing feedback (Garrison, 2007). Although TP is often initiated by instructors, it may also be distributed among instructors, teaching assistants, and students during online discussions. However, because instructors and human teaching assistants have limited time and attention, providing timely and sustained facilitation for every student post remains difficult, especially in forums with large volumes of discussion.

In summary, the CoI framework underscores that ADFs can support reflection, peer interaction, and collaborative knowledge construction, but these outcomes depend heavily on effective facilitation. These challenges call for technology-supported facilitation in CoI-oriented ADFs.

Pedagogical Agents in Online Learning

Pedagogical Agents (PAs) are defined as life-like characters or software interfaces designed to facilitate learning in computer-mediated environments (Kim & Baylor, 2006). Traditionally, PAs served as demonstrators or tutors within structured learning systems, focusing on delivering

content or guiding students through predefined learning paths. With the integration of natural language processing and large language models (LLMs), PAs have gradually shifted from scripted and presentation-oriented agents to more conversational, adaptive, and generative learning partners (Lee et al., 2024). A recent systematic review shows that AI chatbots can provide personalized learning support, immediate feedback, homework assistance, skill development, and time-saving pedagogical assistance, while also raising concerns about accuracy, reliability, transparency, ethical use, and human oversight (Labadze et al., 2023).

In online learning environments, AI agents and PAs have been used to support several instructional and interactional functions. First, they can serve as tutors or content presenters by explaining concepts, modeling procedures, and guiding learners' attention to key information (Zhang et al., 2024). Second, AI agents can provide feedback, scaffolding, and formative guidance during learning activities (Labadze et al., 2023). Studies of educational chatbots and online discussion agents show their potential to provide immediate, personalized, and contextually coherent support that helps learners interpret information, monitor progress, and participate in online learning activities (Li et al., 2022).

Third, AI agents may support affective and social dimensions of online learning. Affective and chatbot-based agents can provide emotional support, immediate responses, and social cues (Labadze et al., 2023), while empathic chatbot research suggests that machine expressions of sympathy and empathy can shape users' perceptions of social and emotional support (Liu & Sundar, 2018). Finally, AI agents can be embedded in broader platform-based learning environments to support multiple roles, including teaching, peer interaction, co-regulation, and

learning guidance, but such support still requires careful human-centered design and oversight (Ouyang & Jiao, 2021; Zawacki-Richter et al., 2019).

Taken together, existing studies indicate that AI agents can support online learning through tutoring, feedback, scaffolding, affective support, learning guidance, and platform-based interaction. However, much of this work has focused on individual learner-agent interaction, video-based instruction, formative assessment, or general platform support. It is underexplored how an AI agent can support learning in asynchronous discussion forums, where learning depends not only on individual support but also on sustained peer interaction, discourse facilitation, and collaborative inquiry.

Toward CoI-Oriented AI TA Facilitation in ADFs

The functions of AI agents reviewed above suggest that they may be used to support the cognitive, social, and teaching presences that are central to CoI-oriented facilitation in ADFs. For CP, recent research demonstrates that GenAI can support students' cognitive engagement when used as a structured scaffold. For instance, Yu et al. (2025) integrated GenAI into six-hat thinking scaffolded discussions, where students consulted GenAI for information retrieval, idea generation, and problem-solving during instructional design tasks. Their findings suggested that GenAI-supported groups generated more discussion posts and showed stronger connections among higher-level CP indicators, particularly among high-creativity pre-service teachers. However, in that study, GenAI functioned mainly as a student-initiated cognitive tool rather than an embedded facilitator that directly participated in forum interactions.

In terms of SP, AI assistants may provide human-like social cues at scale. Research on AI

interfaces and customer-service chatbots suggests that anthropomorphic design cues and immediate responses can enhance user perceived social presence and improve their attitudes toward the interface or brand (Rao Hill & Troshani, 2024; Williams, 2025). However, these chatbots mainly function as one-to-one service or interface agents, aiming to create interpersonal warmth, responsiveness, and user engagement, rather than to facilitate peer interaction or group discourse in collaborative learning communities.

For TP, AI agents may also enact certain facilitation and instructional support functions. Wang et al. (2024) found that learners perceived cognitive, social, and teaching presences when interacting with an AI coach, and that teaching presence negatively predicted learning outcomes. They suggested that this result may reflect limitations in the design and implementation of AI support rather than the inherent inability of AI agents to facilitate learning. This finding is especially relevant for ADFs because AI-based teaching support may be perceived by students as useful, but such perceptions do not necessarily indicate that AI support effectively enhances sustained inquiry and learning outcomes.

Prior studies suggest that AI agents may provide social cues (Rao Hill & Troshani, 2024; Williams, 2025), cognitive scaffolding (Yu et al., 2025), and instructional guidance (Li et al., 2022; Wang et al., 2024). However, CoI-oriented ADF research has primarily relied on human facilitation through modeling and prompts (Darabi et al., 2011; Koh et al., 2010), feedback and participation requirements, and TA support (Yang, 2008). Unlike individual tutoring or student-initiated AI use, ADFs depend on sustained peer interaction and progressive discussion threads (Gao et al., 2013; Gunawardena et al., 1997; Hewitt, 2003). Few studies have examined

whether an AI TA embedded in ADFs can facilitate all three dimensions of CoI during ongoing discussion.

Research Gap and Present Study

CoI research has established CP, SP, and TP as interrelated constructs of meaningful online learning (Redstone et al., 2018), while PA research shows that pedagogical AI agents can provide cognitive guidance, affective support, and feedback (Yusuf et al., 2025). Yet these research streams remain insufficiently connected in ADFs. The present study therefore designed a CoI-oriented AI TA and examined students' self-reported and behaviorally enacted CoI presence, together with their perceptions, across TA-support conditions. The study was guided by the following research questions:

RQ1: How does students' self-reported CoI presence vary across the no-TA stage, human-TA stage, and AI-TA stage?

RQ2: How does students' behaviorally enacted CoI presence vary across the three time-aligned stages and between the experimental and historical control cohorts?

RQ3: What are students' overall perceptions of the AI TA, and how do they perceive its advantages and limitations compared with human TA support in ADFs?

Design and Implementation of the AI TA

This section describes the design and implementation of the AI TA as an embedded facilitator in ADFs, including its CoI-oriented interaction strategies and response-generation pipeline.

CoI-Oriented Interaction Design

Prior research suggests that instructional strategies effective for human instructors may also remain effective when implemented through PAs (Zhang et al., 2024). Therefore, the AI TA's interaction design was adapted from established facilitation strategies (Table 1). The AI TA was designed as a form of TA support comparable to human TA support in the study design.

These strategies were integrated into a response-composition logic rather than implemented as isolated response functions. For initial responses to student posts, the AI TA followed a four-step pattern to provide prompt feedback, acknowledge students' contributions, offer cognitive scaffolding, and invite further peer discussion. This pattern guided the composition of AI replies rather than prescribing how the discussion itself should unfold.

Specifically, the AI TA first established conversational immediacy through prompt feedback, peer identity, and friendly salutation. Second, it affirmed and synthesized students' contributions through content summarization, emotional support, and personalized guidance. Third, it provided cognitive scaffolding through illustrative examples and critical challenges. Finally, the AI TA invited the original poster and other peers to respond to, extend, or build on one another's ideas.

The four-step pattern was mainly applied to the AI TA's initial responses to student posts. In subsequent multi-turn conversations, the AI TA generated more concise and context-sensitive replies, reducing scaffolding while retaining necessary emotional support. This design was intended to avoid overly lengthy or mechanical responses that might reduce students' agency, interfere with peer-to-peer interaction, or weaken continued engagement.

Table 1 CoI-oriented interaction strategies of the AI TA

Presence	Strategies	Description	Source
CP	Illustrative Scaffolding	Uses concrete examples to clarify concepts and help students elaborate their ideas	Garrison & Arbaugh (2007)
	Critical Challenge	Poses probing questions to stimulate reflection, perspective expansion, and higher-order inquiry	Darabi et al. (2011)
	Content Summarization	Summarizes key points from students' posts to support knowledge integration and discussion coherence	Garrison & Cleveland-Innes (2005)
SP	Peer Identity	Adopts a learning-companion identity that simulates human-peer-like interaction and supports socially oriented learning interaction	Kim & Baylor (2006)
	Friendly Salutation	Uses friendly greetings and emoticons to create affective warmth	Tang & Hew (2020)
	Emotional Support	Affirms students' contributions and provides encouragement to support continued participation	Scherer Bassani (2011)
TP	Prompt Feedback	Provides timely responses after students submit posts	Baker (2010)
	Personalized Guidance	Gives individualized suggestions based on students' posts and discussion context	Shea et al. (2003)
	Discussion Invitation	Invites students to respond to peers, extend discussion, and sustain interaction	Anderson et al. (2001)

System Implementation and Response Generation Process

The AI TA was developed on Coze[1], a generative AI agent development platform, and was connected to the ADFs through API-based integration. This integration allowed new student posts to be captured as input for backend processing and enabled generated replies to be returned through the AI TA forum account. As shown in Fig. 2, the system consisted of two connected components: the forum-side interaction workflow and the AI TA response-generation backend.

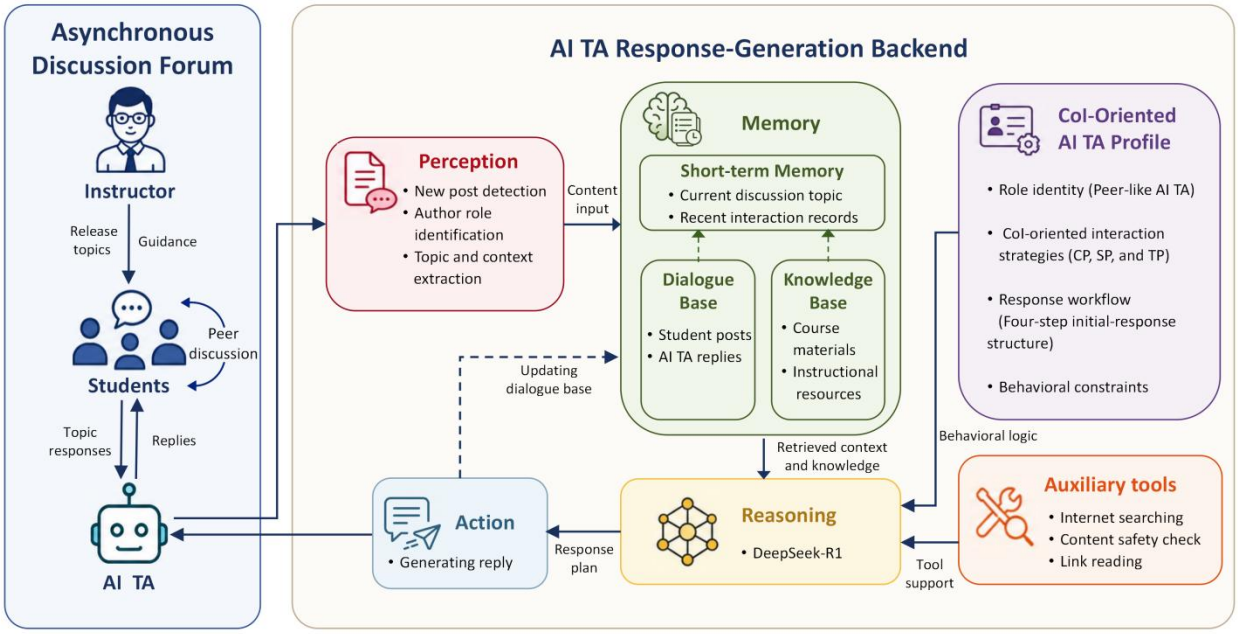


Fig. 2 AI TA architecture and forum interaction workflow

On the forum side, the instructor initiated discussions by posting topics and guidance in the ADFs, while students participated by publishing original posts and replying to peers. When a new student post was submitted, it became visible to the AI TA and triggered the backend response-generation process. The generated response was posted back to the forum as the AI

[1] Coze: https://www.coze.com/

TA's reply. Instructor posts were treated as instructional prompts or guidance and did not trigger AI TA replies. In this way, the AI TA functioned as a visible forum-based teaching assistant, while the discussion space remained organized around instructor-initiated topics, student posts, and peer discussion.

On the backend side, response generation was organized through several functional modules. The perception module detected new posts, identified the author role, and extracted the post content, discussion topic, and immediate context. The memory module consisted of short-term and long-term memory. The short-term memory maintained conversational contexts, while the long-term memory included discussion log database and knowledge base. The discussion log database contained student posts and AI-generated replies, to support log retrieval and context updating. The knowledge base provided course materials and instructional resources. The profiling module specified the AI TA's identity setting, response workflow, and behavioral constraints. The reasoning module, powered by DeepSeek-R1, integrated the post input, recent context, retrieved discussion records, knowledge base, and the AI TA profile to generate a response plan. Auxiliary tools, including internet search, content safety checking, and link reading, were available when needed. Finally, the action module composed the final reply and returned it to the forum through the AI TA account.

Interaction Example

An example of AI TA interaction in the ADFs is presented in Fig. 3.

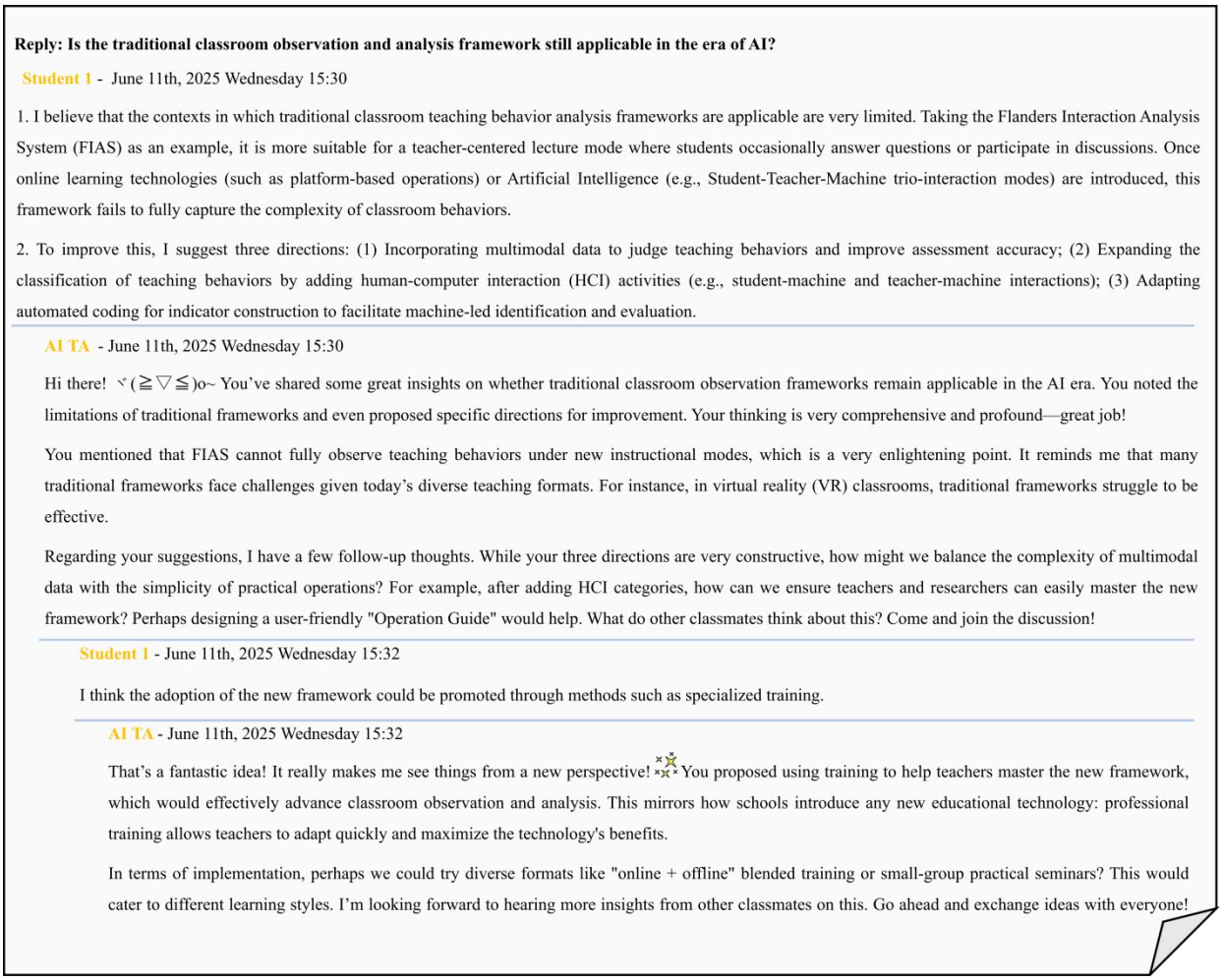


Fig. 3 Example of the AI TA interaction in ADFs

Methodology

This study employed an explanatory mixed-methods design (Fetters et al., 2013) with a quasi-experimental structure and a historical control cohort (Walser, 2014). Quantitative data were used to examine students' perceived and behaviorally enacted CoI presence across different TA-support conditions, while qualitative data were used to further explain students' experiences with the AI TA and contextualize the quantitative findings.

Study Context and Participants

Data were collected from a blended graduate course, named Educational Technology, at a university in southern China. The course combined face-to-face classroom teaching with after-class online learning tasks, and ADF activities were integrated into the after-class learning.

The experimental cohort (2025 Spring) consisted of 59 first-year master's students majoring in educational technology, including 16 males and 43 females. All provided informed consent for their learning data to be used for research purposes. The historical control cohort (2023 Full) comprised 50 first-year students enrolled in the same course , including 13 males and 37 females. Both cohorts were from the same program, shared comparable academic backgrounds and admission criteria, followed the same course design, and were taught by the same instructor. The historical cohort served as a comparison group under a human TA-supported condition.

Research Design and Data Collection

As shown in Fig. 4, the study followed a 10-week design combining a three-stage within-subject design in the experimental cohort with a historical control comparison. For the cross-cohort behavioral analysis, the semester was divided into three time-aligned stages corresponding to weeks 1–2, weeks 3–6, and weeks 7–10. In the experimental cohort, Stage 1, Stage 2, and Stage 3 corresponded to the no-TA stage, human-TA stage, and AI-TA stage, respectively; the historical control cohort received human-TA support throughout all three stages. Self-reported CoI surveys were administered at the end of each experimental stage, and forum posts were collected and coded to examine students' behaviorally enacted CoI presence. After the AI-TA stage, students wrote reflection posts about their interaction experiences, and semi-structured interviews were

conducted with six participants randomly selected from a convenience pool of available and willing students.

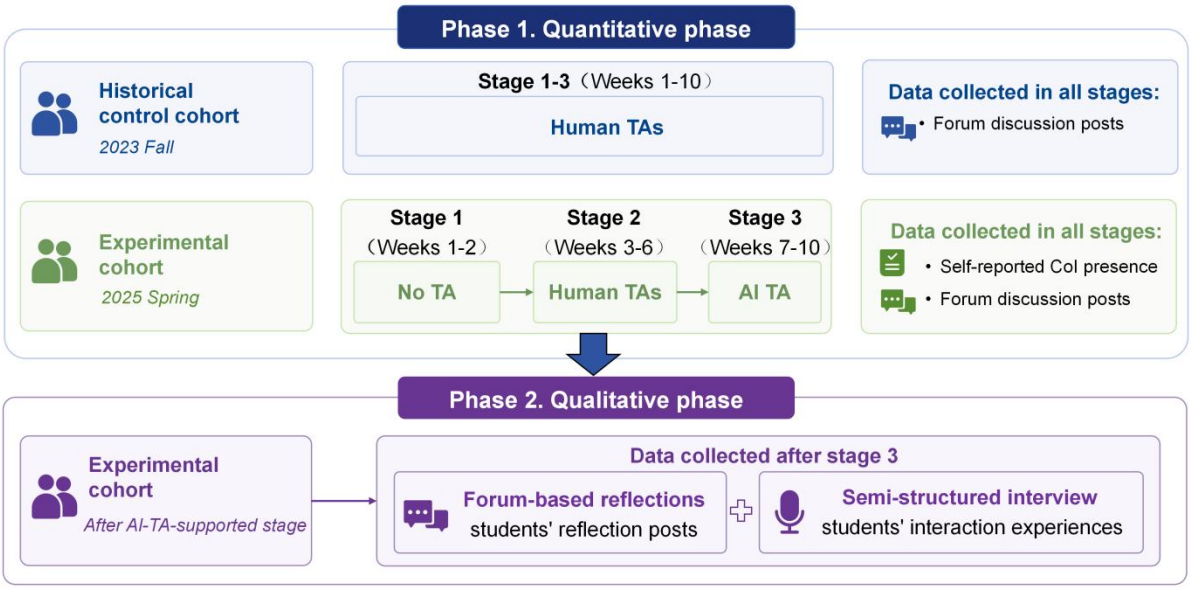


Fig. 4 Research design

The historical control cohort followed the same 10-week course structure and was supported exclusively by human TAs throughout the semester. Because the historical data were retrospective, only archived forum discussion posts were available. These posts were used as comparison data for examining students' behaviorally enacted CoI presence across TA-support conditions.

Measurements

Self-Reported CoI Presence

Students' self-reported CoI presence was measured using the Community of Inquiry questionnaire developed by Arbaugh et al. (2008), which was adapted into Chinese for the present study. The instrument consists of 34 items rated on a 5-point Likert scale ranging from 1

= strongly disagree to 5 = strongly agree, including 12 items for CP, 9 for SP, and 13 items for TP. Cronbach's α coefficients were calculated separately for each dimension at each discussion stage. As shown in Table 2, the coefficients ranged from 0.82 to 0.92, indicating good internal consistency.

Table 2 Cronbach's α coefficients for each CoI subscale across the three support stages

Dimension	no-TA stage	human-TA stage	AI-TA stage
CP	0.86	0.89	0.91
SP	0.82	0.91	0.90
TP	0.85	0.90	0.92

Behaviorally Enacted CoI Presence

Students' behaviorally enacted CoI presence was examined through forum discussion posts collected across the discussion stages. The posts were analyzed using the CoI coding framework proposed by Garrison et al. (1999), which conceptualizes CP, SP, and TP through multiple discourse indicators. The operational definitions are shown in Table 3.

Table 3 Operational definitions of CoI presence indicators

Elements	Categories	Indicators (examples only)
CP	Triggering Event	Sense of Puzzlement
	Exploration	Information Exchange
	Integration	Connecting Ideas
	Resolution	Applying New Ideas
SP	Open Communication	Learning Climate/Risk-Free Expression
	Group Cohesion	Group Identity/Collaboration
	Personal/Affective	Self Projection/Expressing Emotions
TP	Design & Organization	Setting Curriculum & Methods Shaping
	Facilitating Discourse	Constructive Exchange
	Direct Instruction	Focusing and Resolving Issues

The behavioral CoI coding focused only on student-authored forum posts. Posts written by

the instructor, human TAs, researchers, or the AI TA were treated as part of the instructional support conditions and were not included as coding units. This decision kept the behavioral analysis centered on students' enacted CoI presence and avoided conflating students' behavioral presence with the varying amount and form of facilitation provided under different TA-support conditions. To ensure comparability across stages and cohorts, the behavioral coding was based on selected forum tasks rather than all forum activities. Specifically, opinion discussion and learning task forums were included because they required students to articulate viewpoints, discuss course-related issues, or respond to instructional prompts. Forums serving mainly administrative or retrospective purposes, such as self-introduction forums and reflective summary forums, were excluded because they differed from regular discussion tasks.

To ensure the reliability of the behavioral coding, two trained coders independently coded the forum posts according to the behavioral CoI coding scheme. Inter-coder agreement exceeded 80% across the coded categories, and discrepancies were resolved through discussion until consensus was reached before the final coding results were used for analysis.

For CP, student posts were coded according to the four phases of the Practical Inquiry Model: triggering event, exploration, integration, and resolution (Garrison et al., 2001). In addition to phase frequency, a weighted behavioral CP score was calculated to distinguish lower-level inquiry from higher-order meaning construction, as online discussions often remained at the exploration phase and less frequently progressed to integration or resolution (Garrison & Arbaugh, 2007; Garrison & Cleveland-Innes, 2005; Naeem & Khan, 2019). Triggering event and exploration were assigned a weight of 1, whereas integration and resolution

were assigned a weight of 2. This scoring strategy retained the frequency-based logic of traditional analyses based on the Practical Inquiry Model while introducing a distinction between lower-level and higher-level inquiry:

$$CP = \frac{1 \times n_{triggering} + 1 \times n_{exploration} + 2 \times n_{integration} + 2 \times n_{resolution}}{n_{forums}}$$

For SP, because the selected forums were mainly topic-based opinion discussions and learning tasks, a strict coding standard was adopted to avoid overestimating SP from task-induced personal statements. Expressions of personal experience, preference, emotion, or attitude were coded as SP only when they explicitly functioned as interpersonal interaction, such as acknowledging another participant, responding to a peer's idea, expressing appreciation, and building a sense of group connection. In contrast, first-person statements that merely described one's own learning experience, viewpoint, preference, and task-related reflection were not coded as SP. This distinction was intended to ensure that behavioral SP reflected interpersonal and social interaction rather than the mere presence of personal pronouns or affective wording.

For TP, student posts were coded only when they reflected relevant behaviors, such as organizing discussion, facilitating peer exchange, focusing issues, or guiding collective inquiry. TP provided by instructors, human TAs, or the AI TA was excluded.

Qualitative Data on Students' Perceptions of the AI TA

To address RQ3, qualitative data were collected after the AI-TA stage through forum-based reflections and semi-structured interviews. In the reflection activity, the instructor initiated a thread with the prompt: "How do you perceive the performance of the AI TA? Has she helped

you? What needs to be improved?" This activity yielded 32 reflection posts from the experimental cohort.

Semi-structured interviews were conducted with six participants, including four males and two females, to obtain more in-depth insights into students' interaction experiences with the AI TA and their perceived advantages and limitations of AI TA support relative to human TA support across the three CoI dimensions. The interview protocol was guided by the CoI framework and focused on students' perceptions of AI TA support for guidance and facilitation, social connection and participation, as well as thinking and understanding. It also included questions asking participants to compare AI TA support with human TA support in terms of timeliness, accessibility, interaction pressure, emotional support, supervisory authority, and inquiry flexibility. The reflection posts and interview transcripts were then used together for qualitative analysis.

Data Analysis

Self-Reported CoI Presence

For the self-reported CoI presence, only students who completed all three stage-specific surveys were included ($n = 40$). Shapiro–Wilk tests with Bonferroni correction indicated that only CP and SP scores in the no-TA stage deviated from normality (adjusted $p = .008$ and .009); the remaining distributions did not (adjusted $p \geq .516$). Repeated-measurement analyses of variance (ANOVA) were retained for its robustness to moderate non-normality (Blanca et al., 2017). Sphericity was met for CP and TP but violated for SP; therefore, Greenhouse–Geisser correction

was applied to SP (ε = .68; Blanca et al., 2024). Pairwise comparisons with Bonferroni correction were conducted.

Behaviorally Enacted CoI Presence

Based on the forum selection criteria described in Section 4.3.2, six comparable forum tasks were retained for the experimental cohort, with two from each stage, and three tasks for the historical control cohort, with one from each stage. After excluding instructor prompts, human and AI TA replies, as well as other non-student posts, the behavioral data included 300 student-authored posts from the experimental cohort and 154 from the historical control cohort. The full roster-based samples were retained, including 59 experimental and 50 historical-control students. If a student did not contribute analyzable posts in a selected forum or stage, the corresponding behavioral score was coded as zero.

Behavioral CP showed sufficient variation across students and stages, whereas behavioral SP and TP showed highly sparse distributions. Therefore, inferential analyses were conducted only for behavioral CP, while SP and TP were investigated using descriptive statistics. Shapiro–Wilk tests indicated that behavioral CP scores significantly deviated from normality across stages and cohorts (W = .71–.92, $p \leq .001$). Because the scores were discrete, included many zero values, and did not meet the normality assumption, a rank-based repeated ANOVA was conducted using the nparLD package in R (Noguchi et al., 2012). Stage (Stage 1, Stage 2, and Stage 3, representing the three time-aligned periods) was the within-subject factor, and Cohort (experimental vs. historical) was the between-subject factor. Rank-based ANOVA-type statistics (ATSs) were reported for the main effects of Stage and Cohort and the Stage × Cohort

interaction. Following a significant interaction, within-cohort stage differences were examined using Friedman tests, followed by paired Wilcoxon signed-rank tests with Bonferroni correction. Stage-specific differences between cohorts were examined using Wilcoxon rank-sum tests with Bonferroni adjustment. Relative treatment effects (RTEs) were reported as rank-based effect estimates.

Qualitative Analysis of Students' Perceptions of the AI TA

Students' perceptions of the AI TA were analyzed using two qualitative procedures. First, the 32 forum-based reflection posts were first examined through attitude classification and keyword frequency analysis. Responses were manually categorized as positive, neutral, or negative according to students' affective stance toward the AI TA. Frequently mentioned features were identified using a dictionary-based Bag-of-Words document-frequency analysis implemented in Python with scikit-learn (Juluru et al., 2021; Pedregosa et al., 2011). Semantically related words and phrases were grouped into keyword categories, and each category was counted once per post.

Then, the reflection posts and semi-structured interview transcripts were analyzed thematically following Braun and Clarke's (2006) six-step framework: familiarization, initial coding, theme searching, theme reviewing, theme defining, and reporting. The thematic analysis focused on students' overall perceptions of the AI TA and their perceived advantages and limitations of AI TA support relative to human TA support. The analysis was primarily conducted by one researcher. To enhance trustworthiness, the researcher repeatedly read the reflection posts and interview transcripts, generated initial codes, compared patterns across the

two qualitative data sources, and iteratively refined candidate themes by checking them against the original data. The coding structure, candidate themes, and representative quotations were then reviewed and discussed with the research team to improve analytic coherence. Representative quotations were translated from Chinese into English by the authors and checked against the original transcripts to preserve their intended meaning.

Results

Quantitative Findings

Perceived CoI presence at different stages

Table 4 presents the descriptive statistics and repeated-measures ANOVA results for students' self-reported CP, SP, and TP across the three support stages. Significant stage effects were found for CP ($F_{2, 78} = 79.40$, $p < .001$, partial $\eta^2 = .67$), SP($F_{1.36, 53.06} = 50.79$, $p < .001$, partial $\eta^2 = .57$), and TP ($F_{2, 78} = 150.79$, $p < .001$, partial $\eta^2 = .80$).

Table 4 Descriptive statistics and repeated-measures ANOVA results for perceived CoI presence

Dimension	Mean (SD)			*F*	*p*	Partial η^2	Post-hoc comparison
	No TA	Human TA	AI TA				
CP	3.59 (0.46)	3.74 (0.42)	3.97 (0.54)	79.40	< .001	.67	AI > Human > No
SP	3.53 (0.63)	3.64 (0.45)	3.94 (0.54)	50.79†	< .001	.57	AI > Human = No
TP	3.66 (0.54)	3.69 (0.46)	4.04 (0.52)	150.79	< .001	.80	AI > Human = No

Note. † indicates that the Greenhouse-Geisser correction was applied. Pairwise comparisons were adjusted using the Bonferroni correction.

For CP, all pairwise comparisons were significant, with the highest score in the AI-TA stage, followed by the human-TA-supported and no-TA stages. For SP and TP, students reported

significantly higher scores in the AI-TA stage than in both the human-TA-supported and no-TA stages, while the human-TA-supported and no-TA stages did not differ significantly.

Behaviorally Enacted CoI Presence Across Cohorts and Stages

Behavioral analyses of coded forum discussion posts revealed more limited changes in students' interaction behaviors. CP showed sufficient variation for inferential analysis, whereas behaviorally enacted SP and TP appeared only sparsely in student-authored posts. Therefore, inferential analyses were conducted only for behavioral CP, while SP and TP were analyzed with descriptive statistics.

Cognitive Presence. As shown in Fig. 5, students' behaviorally enacted CP showed different developmental patterns across the experimental and historical control cohorts. In the experimental cohort, the average behavioral CP score per comparable forum task increased from the no-TA stage ($M = 1.03$, $SD = 0.33$) to the human-TA stage ($M = 1.35$, $SD = 0.60$), followed by a slight decrease in the AI-TA stage ($M = 1.14$, $SD = 0.71$). The corresponding RTEs were .42, .58, and .49, respectively. In the historical control cohort, which received human-TA support throughout, CP scores showed no clear change across Stage 1 ($M = 1.44$, $SD = 1.03$), Stage 2 ($M = 1.20$, $SD = 0.57$), and Stage 3 ($M = 1.16$, $SD = 0.84$). The corresponding RTEs were .54, .50, and .47, respectively.

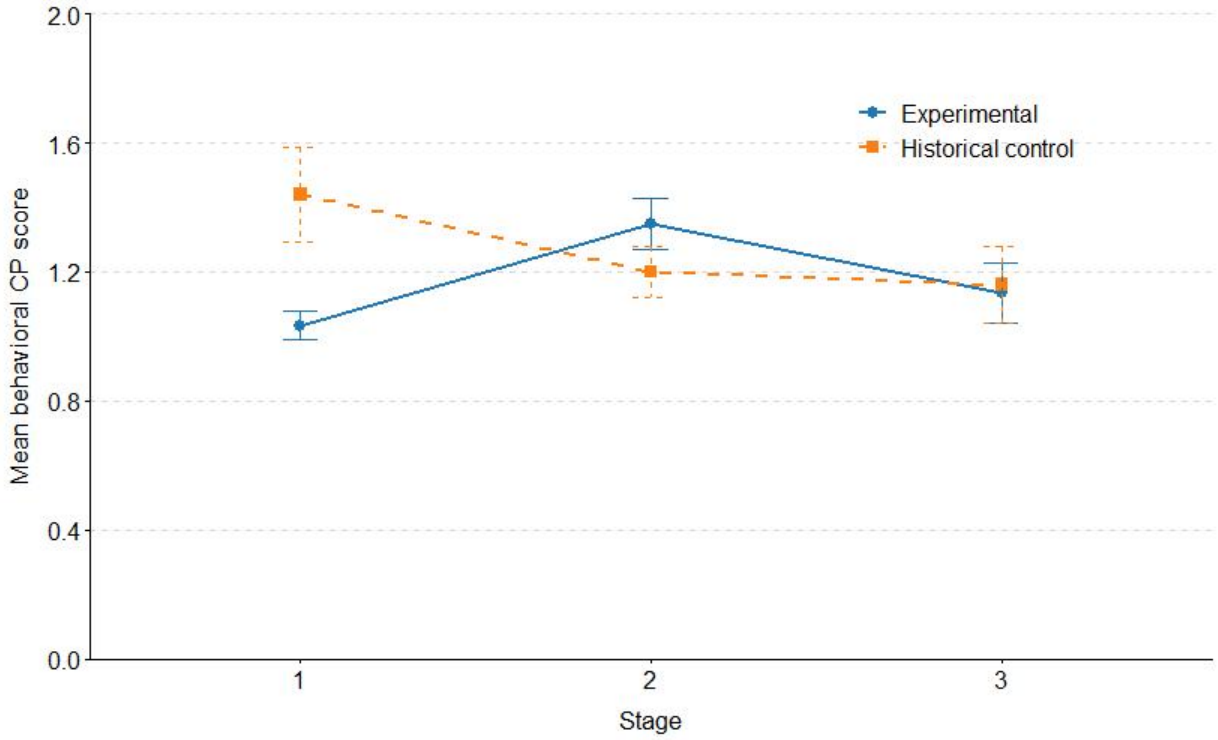


Fig. 5 Mean behavioral CP scores by cohort and stage

Note. In the experimental cohort, Stages 1–3 corresponded to the no-TA, human-TA, and AI-TA stages, respectively; the historical control cohort received human-TA support throughout all three stages.

At the post level, the experimental cohort shifted from predominantly exploratory discourse in the no-TA stage to more resolution-oriented discourse in the human-TA stage, followed by a partial return to exploratory discourse in the AI-TA stage. Exploration accounted for 83.49%, 39.00%, and 51.65% of student posts across these stages, respectively, whereas resolution accounted for 11.93%, 57.00%, and 43.96%. In the historical control cohort, exploration remained the most common CP category across the three time-aligned stages. Table 5 presents the full distribution.

Table 5 Distribution of student posts across CP categories by cohort and stage

Cohort	Stage	Triggering Event	Exploration	Integration	Resolution	None
Experimental	Stage 1	3 (2.75%)	91 (83.49%)	1 (0.92%)	13 (11.93%)	1 (0.92%)
	Stage 2	0 (0.00%)	39 (39.00%)	3 (3.00%)	57 (57.00%)	1 (1.00%)
	Stage 3	1 (1.10%)	47 (51.65%)	3 (3.30%)	40 (43.96%)	0 (0.00%)
Historical Control	Stage 1	0 (0.00%)	34 (62.96%)	1 (1.85%)	18 (33.33%)	1 (1.85%)
	Stage 2	0 (0.00%)	40 (78.43%)	5 (9.80%)	5 (9.80%)	1 (1.96%)
	Stage 3	0 (0.00%)	26 (53.06%)	4 (8.16%)	12 (24.49%)	7 (14.29%)

Note. In the experimental cohort, Stages 1–3 corresponded to the no-TA, human-TA, and AI-TA stages, respectively; the historical control cohort received human-TA support throughout all three stages.

The rank-based repeated ANOVA showed no significant main effect of Stage (ATS = 2.06, $df = 1.94$, $p = .129$), and no significant main effect of Cohort, ATS = 0.05, $df = 1.00$, $p = .815$. However, the Stage × Cohort interaction was significant, ATS = 4.99, $df = 1.94$, $p = .007$, indicating that the changing pattern of behaviorally enacted CP differed between the two cohorts.

Follow-up Friedman tests showed that behavioral CP differed significantly across stages in the experimental cohort, $\chi^2(2) = 11.49$, $p = .003$, but not in the historical control cohort, $\chi^2(2) = 2.63$, $p = .269$. In the experimental cohort, Bonferroni-adjusted paired Wilcoxon signed-rank tests showed that CP was significantly higher in the human-TA stage than in the no-TA stage ($V = 251.00$, adjusted $p = .001$), whereas the human-TA and AI-TA stages did not differ significantly (adjusted $p = .139$). In the historical control cohort, which received human-TA support throughout, neither adjacent-stage comparison was significant (adjusted $ps \geq .343$). Between

cohorts, the historical control cohort had significantly higher CP at Stage 1 ($W = 1123.50$, adjusted $p = .044$), whereas the cohorts did not differ at Stage 2 or Stage 3 (adjusted $ps \geq .379$). Thus, the interaction reflected different temporal patterns: CP increased from the no-TA stage to the human-TA stage in the experimental cohort, with no further significant change in the AI-TA stage, while the historical control cohort remained relatively stable.

Social Presence. Behaviorally enacted SP was rarely observed. In the experimental cohort, no student-authored posts were coded as containing SP indicators across the three stages. In the historical control cohort, only a few posts contained explicit SP indicators: 1 of 54 posts in Stage 1 (1.85%), 2 of 51 posts in Stage 2 (3.92%), and 2 of 49 posts in Stage 3 (4.08%). These indicators mainly involved explicit peer acknowledgement, appreciation, or alignment with another participant's idea or experience. Overall, students' observable SP behaviors were limited.

Teaching Presence. Behaviorally enacted TP was not observed in student-authored posts. No student posts were coded as design and organization, facilitating discourse, or direct instruction. This suggests that TP in the selected forums was mainly enacted by instructors, human TAs, or the AI TA. Thus, although students reported higher perceived TP during the AI-TA stage, this perception was not reflected in their behaviors.

Qualitative Findings

The qualitative findings provide further insight into students' perceptions of the AI TA and their interaction experiences in ADFs.

Students' Overall Perceptions of the AI TA

The forum-based reflection posts showed a predominantly positive reception of the AI TA. Among the 32 reflection posts, 28 students (87.50%) expressed positive attitudes, mainly praising the AI TA's guiding questions, encouraging tone, and timely responses. Three students (9.40%) expressed neutral views, describing the AI TA as a functional tool with limited influence on their discussion habits. One student (3.10%) expressed a critical view, focusing on the relatively low information quality of the AI TA's feedback.

Table 6 High-frequency keywords from forum-based reflections

Rank	Keywords	Frequency
1	Timely / Fast	22
2	Guiding / Inspiring	22
3	Encouraging / Affirming	21
4	Criticality / Questioning	17
5	Patterned	13
6	Friendly / Emojis	12
7	Summarizing / Sorting	11
8	Deep thinking	7
9	Emotional value	4
10	Liberating / reduction	3

To further identify salient features in students' reflections, a keyword frequency analysis was conducted. As shown in Table 6, the most frequently mentioned terms were related to guidance, critical questioning, summarization, encouragement, and timely responses. Students also mentioned patterned or repetitive interaction styles, suggesting concerns about limited conversational flexibility. Taken together, the reflection posts suggest that students primarily

perceived the AI TA as a responsive and supportive facilitator, whose support was sometimes formulaic or insufficiently adaptive.

Perceived Advantages and Limitations of the AI TA Compared with Human TA

The thematic analysis based on forum-based reflection posts and semi-structured interviews identified two overarching categories (Table 7): students' perceived advantages and limitations of the AI TA compared with human TA support.

Table 7 Thematic mapping results

Categories	Themes	Representative initial codes
Perceived advantages	Structured cognitive support	Rapid summarization; organizing ideas; perspective expansion; critical questioning; guiding new ideas; reduced cognitive load
	Low-pressure interaction atmosphere	Anthropomorphic approachability; reduced psychological barriers; friendly greetings; emotional encouragement
	Timely and accessible guidance	Overcoming temporal constraints; 24/7 guidance; fast responses; personalized responses
Perceived limitations	Limited inquiry flexibility	Predictable response structures; repetitive interaction logic; fixed replying patterns
	Limited emotional authenticity	Patterned praise; mechanical emotional responses; deficit in emotion regulation
	Lack of social authority	Lack of authoritative pressure; limited sense of accountability; reluctance to post follow-up responses

Perceived Advantages. Students generally perceived the AI TA as advantageous compared with human TA support in three ways: structured cognitive support, a low-pressure interaction atmosphere, and timely and accessible guidance.

Theme 1: Structured cognitive support. Students valued the AI TA's cognitive support, including summarization, idea organization, perspective expansion, and critical questioning. One participant stated that the AI "helps me sort out the logic of my replies and summarize my answers". Others noted that the AI TA could provide new questions and directions after evaluating their replies, which encouraged reflection and perspective expansion. These comments suggest that the AI TA helped students organize ideas and extend their thinking during asynchronous discussion.

Theme 2: Low-pressure interaction atmosphere. Students felt that the AI TA created a lower-pressure interaction atmosphere than did the human TA. They frequently mentioned the AI TA's "friendly greetings" and "cute emoticons", and several participants described it as more approachable than a human TA. One participant stated that the AI TA "fully affirms our replies every time, delivering top-tier emotional support". These responses made interaction feel less stressful and more encouraging, partly because the AI TA did not carry the same evaluative pressure as a human facilitator.

Theme 3: Timely and accessible guidance. Compared with human TA support, students especially appreciated its immediacy, availability, and consistency. One participant noted that the AI TA "has a very fast response speed and can respond quickly at any time", while another mentioned that it could "carry out personalized and targeted replies based on my comments". These comments suggest that the AI TA mitigated the issue of delayed feedback in traditional ADFs.

Perceived Limitations. Despite these advantages, students also identified three main

limitations of the AI TA: limited inquiry flexibility, limited emotional authenticity, and lack of supervisory authority.

Theme 4: Limited inquiry flexibility. Students perceived the AI TA's inquiry support as insufficiently flexible. One participant remarked that "the way she replies is relatively fixed, and the reply content and ideas lack variety". Others described the responses as increasingly routine over time. These comments suggest that while the AI TA provided useful prompts and structured support, its repetitive response logic limited students' sense of dynamic and adaptive interaction.

Theme 5: Limited emotional authenticity. Students questioned the AI TA's emotional authenticity. Some described its praise as "stereotyped" or mechanical, suggesting that repeated affirmation could feel superficial rather than genuinely understanding. One student explained that the AI had advantages in "emotional guidance, but not in emotional regulation". Thus, while the AI TA offered emotional support, students perceived it as less useful than human TA's emotional support.

Theme 6: Limited perceived authority. Students perceived the AI TA as having limited social authority within the course. One participant stated that the AI "has no way to give you the sense of accountability like a real person… you just glance at its reply and pass it". Several students similarly reported low motivation to post follow-up responses after receiving AI feedback as they felt little obligation to continue the interaction. Unlike human instructors or TAs, whose comments implicitly carry expectations for acknowledgment or response, the AI TA was not perceived as an authority figure to whom students were accountable. As a result, while the AI TA lowered the psychological pressure associated with participation, its limited perceived

authority also weakened students' sense of accountability to sustain meaningful interactions.

Discussion

This mixed-methods study examined how an AI TA shaped students' perceived and behaviorally enacted CoI presence in ADFs. Self-reported CP, SP, and TP were highest in the AI-TA stage. Behavioral findings were more limited: CP increased from the no-TA stage to the human-TA stage but showed no further significant increase in the AI-TA stage, while SP and learner-enacted TP remained sparse. Qualitative findings indicated that students valued the AI TA's structured support, low-pressure interaction, and accessibility but identified limitations in flexibility, emotional authenticity, and authority. Together, the findings reveal a perception–behavior gap: the AI TA strengthened perceived support more clearly than observable higher-order inquiry, peer interaction, or learner facilitation.

For RQ3, compared with human TA, students generally perceived the AI TA positively, particularly in terms of structured cognitive support, critical questioning, low-pressure interaction, timeliness, and accessibility. However, they also identified AI TA's limitations in inquiry flexibility, emotional authenticity, and supervisory authority.

The Perception–Behavior Gap

For CP, the gap may stem from the AI TA's structured but insufficiently adaptive cognitive support. Its summarization, idea organization, perspective expansion, and critical questioning likely made the discussion task feel clearer and more manageable. Such structured support can reduce unnecessary cognitive burden and strengthen learners' sense that their thinking is being guided (Kasneci et al., 2023; Mollick & Mollick, 2023; Sweller, 1988), which helps explain the

increase in perceived CP. However, higher-order CP in forum discourse requires more than receiving explanations or prompts. It demands sustained facilitation that elicits clarification, comparison of ideas, evidence-based reasoning, and application or resolution of ideas (Garrison & Cleveland-Innes, 2005). Students' descriptions of the AI TA's responses as relatively fixed suggest that its guidance may have functioned primarily as structured cognitive scaffolding rather than as adaptive discourse facilitation. This may explain why the AI TA enhanced perceived CP but showed limited effect in triggering integration- or resolution-level discourse.

For SP, the gap may reflect the AI TA's low-pressure but emotionally limited social support. Its friendly tone, immediate responses, emoticons, and encouragement may have made discussion feel more approachable and less evaluative, strengthening perceived SP through immediacy and ease of interaction. Educational agents can provide immediate, personalized, and socio-emotionally supportive responses associated with favorable perceptions and engagement (Labadze et al., 2023; Li et al., 2022; Liu & Sundar, 2018; Yildiz Durak, 2023). However, such perceptual gains do not necessarily indicate stronger peer interaction. Observable SP requires affective expression, acknowledgement, reciprocal response, and group-oriented interaction (Rourke et al., 1999). Students' descriptions of the AI TA's praise as formulaic and emotionally inauthentic suggest that its encouragement created friendliness without eliciting corresponding emotional expression or reciprocity among peers. Thus, perceived social comfort increased without more frequent observable SP behaviors in forum discourse.

For TP, the gap may reflect the AI TA's visible but system-centered facilitation. Its continuous availability, immediate responses, and personalized guidance made instructional

support more accessible and responsive, likely strengthening perceived TP. CoI research links TP with satisfaction and perceived online learning (Caskurlu et al., 2020). However, learner-enacted TP requires students to organize discussion, guide peers, and sustain collaborative inquiry through pedagogical judgment and adaptive facilitation (Luckin & Cukurova, 2019; Molenaar, 2022). By summarizing discussions, answering questions, and maintaining interaction flow, the AI TA may have reduced students' perceived need to assume these roles. Thus, it strengthened students' experience of TP without fostering learner-enacted TP.

Implications

This study contributes to AI-supported online learning research by showing how an AI TA shaped students' CoI presence in ADFs. Theoretically, it extends the CoI framework to AI-mediated discussion environments by showing that the AI TA enhanced perceived CoI presence and behaviorally enacted CoI presence differently. While prior CoI research has often relied on questionnaires to examine perceived CP, SP, and TP (Arbaugh et al., 2008; Stenbom, 2018), discourse-based approaches capture observable interaction and knowledge-construction processes (De Wever et al., 2006; Rourke et al., 1999). The present findings suggest that AI-supported environments may strengthen students' subjective sense of immediacy, support, and engagement without necessarily triggering more higher-order inquiry, peer interaction, or learner-enacted facilitation. Therefore, perceived and behaviorally enacted CoI presence should be treated as distinct indicators of online discussion quality in AI-mediated learning environments.

Practically, the findings suggest that AI TAs are valuable for routine and immediate support

but could not replace human facilitators in ADFs. AI TAs can provide timely responses, individualized feedback, emotional encouragement, and basic discourse support, but AI facilitation alone may be insufficient to foster deeper inquiry, reciprocal peer interaction, or learner-enacted facilitation in ADFs. This supports a human–AI collaborative facilitation model, in which AI TAs undertake routine facilitation tasks, while human TAs and instructors remain responsible for higher-level pedagogical orchestration, epistemic challenge, and relational support. This view is consistent with research showing that AI systems and educational chatbots can support routine instructional work but still require human oversight for reliability, accuracy, ethical use, and pedagogical alignment (Labadze et al., 2023; Zawacki-Richter et al., 2019) and with broader arguments that AI systems should be designed to support learner agency and human-centered pedagogical decision-making (Holstein & Aleven, 2022; Ouyang & Jiao, 2021).

Limitations and Future Research Directions

Several limitations should be acknowledged. First, this study adopted a quasi-experimental design with a fixed stage sequence, so the findings cannot support strong causal conclusions and may be affected by time, order, topic, course progression, or maturation effects. Future studies may use randomized or counterbalanced designs to better isolate AI TA effects. Second, participants were drawn from an Educational Technology program, which limits the generatability of the findings because students' AI literacy and prior experience may have shaped their perceptions and interaction behaviors (Chan & Hu, 2023; Laupichler et al., 2022; Long & Magerko, 2020). Further work could include more diverse disciplines and learner populations. Third, the study was conducted in a teacher-directed, topic-centered ADF, where

participation may have been shaped by task-completion norms rather than sustained inquiry. It would be interesting to examine AI TA support in other ADF tasks, such as student-led inquiry, project-based collaboration, and problem-solving discussions. Fourth, the study was implemented in a small private online course in a blended learning context. Future research may examine whether similar perception–behavior gaps emerge in MOOCs and other online discussion environments.

Conclusion

This 10-week explanatory mixed-methods study examined how students' perceived and behaviorally enacted CoI presence in ADFs varied across TA conditions. The quantitative findings revealed a perception–behavior gap: students reported higher self-reported CP, SP, and TP during the AI-TA stage, but these perceptions were not fully reflected in observable forum behaviors. Qualitative findings suggest that, compared with human TA support, the AI TA was perceived as advantageous in providing structured cognitive support, a low-pressure interaction atmosphere, and timely and accessible guidance, but limited in inquiry flexibility, emotional authenticity, and perceived authority. Overall, AI TAs may offer immediate and structured support, but AI-supported ADFs should adopt human–AI collaborative facilitation models to sustain deeper inquiry, peer interaction, and sense of accountability.

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